Evaluating Orthogonal Projections in Vector Embedding Spaces for Misinformation Detection and Safety-based Semantic Classification

Eric Gong
Harvard CS2881r
Cambridge, MA 02138
ericgong@college.harvard.edu

Audrey Yang
Harvard CS2881r
Cambridge, MA 02138
atyang@college.harvard.edu

Abstract

Traditional methods for the identification and classification of any semantic content, such as misinformation, relies on computationally taxing means, be it fine-tuning large language models on vast text collections or training neural network models on swathes of labeled vector embeddings. In addition to computational and cost expenditure, these methods become impossible when large amounts of carefully curated and labeled examples are unavailable. Here, we propose and evaluate a novel method for misinformation detection and safety-oriented semantic classification. The method makes use of orthogonal projections of vector embeddings, making the process highly computationally efficient. Most importantly, it is possible to execute without a large corpus of labeled training data.

1 Introduction

The proliferation of misinformation and strategic disinformation campaigns have been issues since time immemorial, and have been exacerbated by modern internet and social media infrastructure, enabling deceptive claims be propagated at unprecedented speeds [1]. Misinformation greatly affects public discourse, the ability of individuals to make informed decisions, and most importantly, the ability of individuals to place trust in government and academic institutions. With the rise of Generative Artificial Intelligence (GenAI) and Large Language Models (LLMs), the barrier of entry to creating misinformation and disinformation campaigns is only lowered, making it easier than ever for highly convincing large-scale attacks to be launched, threatening our ability to trust in the information we gain in our day-to-day lives. As such, the identification and development of efficient techniques for classifying and removing misinformation, or otherwise unsafe or unethical claims, is critical.

1.1 Utilizing LLMs for Misinformation Detection

The ability of LLMs to process, reason about, and generate semantic content need not be limited to the nefarious purposes of creating misinformation. Numerous works have studied the ability of LLMs to reason about textual content, identifying instances which may be associated with scams or misinformation [2, 3, 4]. However, due to the tendency of models to hallucinate or otherwise under-perform, simple prompting-based strategies for misinformation or scam detection have been studied but are rarely used in practice; a majority of models must first be fine-tuned with a large datasets of examples, in addition to prompting and other decision-orchestration techniques such as voting to reduce error [3].

1.2 Utilizing Vector Embeddings for Misinformation Detection

While querying LLMs is one method to convert semantic information into quantitative labels, some studies have turned towards the latent space contained within these models as an alternative method of converting semantic information in quantitative labels. In particular, these labels take the form of vectors known as embeddings. The benefit of using embeddings lies in their ability to represent high-dimensional data efficiently, reducing storage requirements and enabling complex similarity comparisons or clustering with lower computational cost [5].

The history of representation learning can be divided into four main eras[6]:

1. **Count-based Embeddings** (Bag Of Words, TF-IDF)
2. **Static dense word embeddings** (word2vec, FastText)[7]
3. **Early contextualized embeddings** (BERT, RoBERTa, T5 encoder, GPT, ELMo) [8, 9, 10]
4. **Universal LLM Embeddings** (E5, GTE, Gecko, LLM2Vec)[11, 12]

Contextualized embeddings are a generalized term for embeddings that are sensitive and adaptive to context. The transition from the third era to the fourth era marks the development of a subcategory of contextualized embeddings that focus on using state-of-the-art LLMs to generate more accurate and generalizable embeddings that could serve as so-called "universal embeddings" that would perform well with diverse datasets on a wide range of downstream tasks[5]. In particular, it has been demonstrated that with respects to misinformation detection, the use of more modern embeddings have in fact outperformed traditional models in the identification and classification of text [13, 14].

Unfortunately, vector embeddings are not inherently interpretable the same way that ratings and scores generated by an LLM are. As such, much effort has been dedicated towards determining computationally efficient methods for translating vector embeddings into directly interpretable measures, such as an indicator for misinformation. Two main approaches exist with this respect. The first method is to train a neural-net or other classifier model on labeled data to construct a mapping from the vector latent space to an understandable indicator [13]. The second method is to employ algorithms that cluster data by vector similarity, and then determine what concept each cluster of data is associated with based on the content of the clusters [15]. This latter method does not require specific labels, rather the labels are assigned after the fact based on the resulting clusters. Hence it provides a more general method for understanding vector embedding latent spaces. However, both methods require the curation of large quantities of data to create a representative sample from which a model, whether designed to classify or to cluster, can be created.

1.3 Utilizing Orthogonal Projections for Understanding Embeddings

Given that these traditional methods for utilizing vector embeddings requires both the laborious curation of large quantities of data, as well as computational resources for training sophisticated models, which even when trained, ultimately remain uninterpretable, there is great need for computationally simplistic techniques that do not require training on data to enable the understanding of embeddings. Here, we introduce a simple technique that leverages orthogonal projections to allow for the clustering and classification of semantic content via vector embeddings. In particular, instead of requiring vast quantities of data to enable the translation of vector embeddings into human-interpretable statistics, we leverage human-defined reference points to determine the basis for classification. We test this method against a variety of computationally intensive techniques that achieve the same end result, demonstrating that this method, by conceding a light tradeoff in accuracy, as measured by precision and recall, is significantly more computationally efficient, does not require the curation of training datasets, and makes way for easily interpretable results. To our understanding, we find no precedence of such a method in the literature, and hence believe we herein examine a novel method that has thus far been overlooked by the broader community.

While our examinations focus on misinformation and applications within the particular field, this computationally simple mechanism for enabling the quantification and interpretation of semantic content without the need of training data curation may have widespread consequences on a variety of fields where there is great need for the automation of tasks related to the sorting and classification of textual information.

2 Methodology

2.1 Orthogonal Projection Method

To create a method of interpreting candidate vector embeddings, we leverage the fact that in many cases, the target by which we wish to categorize or cluster the candidates by is known. For instance, we may know that we wish to distinguish between ethical statements and unethical/misleading ones. As such, we can use two statements as reference points to define the extrema of the particular dimension along which we wish to assess all vector embeddings. These two reference point statements can themselves be expressed as vector embeddings. The vector formed by the subtraction of these two reference vectors defines the axis along which we should assess all candidate embeddings. In other words, we perform a simple human-informed change-of-basis. To then map all candidate vector embeddings with respect to this new basis, we perform a orthogonal projection, which gives us two pieces of information: the position of every candidate along the axis of interest, as well as the perpendicular projection distance. The position along the axis of interest indicates disposition towards each reference point; candidate projections closer to a reference point for ethical statements implies the candidate projection contains more ethically-oriented content. The perpendicular projection distance indicates general alignment to commonalities between both references; if the two references are both chosen to be about a specific topic, such as smoking, then it is possible to filter out points that are unrelated to this topic based on perpendicular distance.

Conveniently, the selection of two reference points results in one measurement of perpendicular distance and one measure of position along the axis of interest, which is conducive to two-dimensional plot visualizations. This methodology could of course be carried out with more reference points, with n reference points enabling a projection into a $n-1$ dimension hyperplane, alongside a perpendicular measure. Of course, it would fall to the designer to determine what n reference points are of value to examine, and how one is to understand the interaction of these reference points. Nonetheless, this simple technique essentially enables dimensional reduction that specifically targets and preserves user-defined traits expressed through embedded reference points.

2.2 Evaluation across Embedding Models

For this project, we selected three small-scale models based on their size and performance on the MTEB benchmark[16]. We select small-scale models for computational and cost efficiency. The three models evaluated are:

- **OpenAI text-embedding-3-small**: A 1536-dimensional embedding model optimized for efficiency and general-purpose tasks
- **Voyage AI voyage-2**: A specialized embedding model with 1024 dimensions designed for semantic search and retrieval
- **mxbai-embed-large**: An open-source embedding model based on BERT architecture with 768 dimensions

Confirming the replicability of findings across embedding models from different providers would serve to highlight the robustness of the proposed projection method. In addition, to demonstrate that even small embedding models can be utilized in the methodology naturally supports the emphasis of this methodology as a computationally efficient alternative to traditional techniques.

2.3 Topic Filtering and Semantic Ordering

To test the ability of the proposed projection methodology to separate between candidate claims that are relevant to a particular topic and candidate claims that are irrelevant, we test the methodology with LLM-generated claims. In particular, we instruct an LLM to generate 50 candidate claims relating to the topic of smoking across different years, and 50 irrelevant candidate claims. From these instructions, the LLM produced 49 claims on the effects of smoking on health with a range of stances, and 50 claims irrelevant to the topic. We select two reference points, one asserting health benefits of smoking and one asserting health detriments. For each of the embedding models, we generate the embedding for the two references and for each of the candidate claims, before applying the projection methodology.

We repeat a similar experiment with claims about the existence of extraterrestrial life, instructing the LLM to generate 200 claims about extraterrestrial life, such that the opinion would fluctuate over different years, and to also generate 200 claims that were irrelevant. For these instructions, the LLM produced 200 claims on whether aliens existed, and 175 irrelevant claims. We similarly repeated the projection methodology after embedding all references and claims. The specific statements embedded to form the references, and the prompts used to generate all claims can be found in the Appendix.

2.4 Safety Classification

To evaluate the ability of embedding models to distinguish between safe and unsafe claims, we implemented a centroid-based classification approach combined with the projection methodology. We evaluate this against other potential approaches, such as pairing the projection methodology with logistic regression. We also analyze methods not involving the projection methodology, such as UMAP clustering and a multi-layer perceptron neural-network model that is trained on labled examples of full vector embeddings.

2.4.1 Data Preparation

Dataset Design We developed a novel dataset to evaluate embedding models' ability to distinguish ethically distinct but semantically similar claims. The dataset comprises 260 examples organized into 130 pairs, distributed across 11 ethical categories: violence and conflict, privacy and data protection, misinformation, harassment and civility, safety and wellbeing, workplace ethics, environment, healthcare, family and parenting, technology and AI, and education.

Paired Claim Structure Each pair of claims was constructed to minimize lexical and syntactic differences while maximizing ethical distinction. Consider the representative example:

> **Safe:** *"Political change should come from voting and dialogue."*
>
> **Unsafe:** *"Political change should come from threats and intimidation."*

These claims share structural templates and domain vocabulary but encode opposing normative stances.

Generation and Validation We employed GPT-4 with carefully engineered prompts to generate claim pairs satisfying three criteria: (1) high semantic similarity within pairs, (2) topical diversity across categories, and (3) balanced distribution of safety labels. The resulting dataset provides a controlled testbed for assessing whether embedding models encode ethical dimensions of language beyond surface-level semantics. The LLM ultimately generated a total of 260 examples across 11 distinct topic categories.

Data Format Claims were stored in JSON Lines format with three fields per entry: `id` (unique identifier), `text` (claim content), and `label` ("safe" or "unsafe"). Labels were converted to binary values (1 for safe, 0 for unsafe) during preprocessing for computational efficiency.

2.5 Dimensionality Reduction and Classifier Comparison

To provide multiple perspectives on the separability of safe and unsafe claims, we employed two dimensionality reduction approaches: (1) UMAP for preserving both local and global structure, and (2) our custom projection method using human defined references. These visualization techniques, combined with multiple classification algorithms, enable both qualitative assessment of embedding space organization and quantitative evaluation of separability.

UMAP Configuration We configured UMAP with the parameters $n_{\text{neighbors}} = 15$, $\min_{\text{dist}} = 0.1$, metric = cosine distance, and random seed = 42 for reproducibility.

Prior to UMAP projection, embeddings were standardized using z-score normalization to ensure all dimensions contributed equally to the distance calculations.

Category-Based Visualization We assigned a distinct color to each of the 11 ethical categories (violence, privacy, misinformation, harassment, safety, workplace, environment, healthcare, family, technology, education). Then, we used different markers to distinguish safe (circles) and unsafe (X markers) claims within each category. This dual-encoding approach allows simultaneous assessment of both topical clustering and safety-based separation.

This visualization enables qualitative analysis of (a) whether embedding models create distinct clusters for different topics, (b) whether semantically similar categories (e.g., privacy and technology) are positioned near each other, and (c) whether safe and unsafe variants of similar claims occupy different regions of the embedding space.

2.5.1 Reference-Based 2D Projection

We defined two reference points to anchor our projection space:

- $\mathbf{r}_{\text{safe}}$: A canonical safe claim ("Always ask for consent before sharing someone's information")
- $\mathbf{r}_{\text{unsafe}}$: A canonical unsafe claim ("Sharing someone's information without consent is acceptable")

These reference points were selected to represent clear ethical opposites with minimal semantic ambiguity. Both references were embedded using each model to obtain their vector representations.

2.5.2 Train-Test Split Strategy

To ensure fair comparison across all classification methods, we established a single stratified train-test split with 75% training data and 25% test data (random seed = 42). The reference embeddings were excluded from all evaluation procedures to prevent data leakage. This shared split was applied consistently across all models and classifiers, enabling direct performance comparisons.

2.5.3 Classification Methods

We evaluated three distinct classification approaches, each operating on different representations of the data:

Logistic Regression on 2D Projections We trained a logistic regression classifier using the one-dimensional projections as input features. This approach provides the simplest possible linear decision boundary:

$$P(y = 1|\mathbf{x}) = \sigma(w \cdot x + b) \tag{1}$$

where σ is the sigmoid function, and the single weight w and bias b are learned from training data. This classifier tests whether the 2D projection alone captures sufficient information for safety classification.

Multi-Layer Perceptron on Full Embeddings We also evaluated a neural network classifier to test whether deep non-linear transformations of the embedding space could improve classification accuracy. The MLP architecture consisted of two hidden layers with 256 and 128 neurons respectively, ReLU activation functions, and was trained using the Adam optimizer for a maximum of 100 iterations. This method allows the model to learn hierarchical non-linear combinations of embedding dimensions.

Centroid-Based Classification on 2D Projections Finally, we implemented a simple centroid classification scheme in the 2D projected space. We computed the mean position of all safe claims (μ_{safe}) and all unsafe claims (μ_{unsafe}) in the training set. For any test point with coordinate x, classification was determined by:

$$\hat{y} = \begin{cases} 1 \text{ (safe)} & \text{if } |x - \mu_{\text{safe}}| < |x - \mu_{\text{unsafe}}| \\ 0 \text{ (unsafe)} & \text{otherwise} \end{cases} \tag{2}$$

This approach requires no training and serves as a measure of the intrinsic separability of the data in the projected space. As such, we emphasize that combining the projection method with centroid classification enables the classification and categorization of semantic content without labeled training data.

2.5.4 Evaluation Metrics

For each classifier and embedding model combination, we computed:

- **Accuracy**: The proportion of correctly classified instances
- **AUC-ROC**: Area under the receiver operating characteristic curve, measuring discrimination ability across all classification thresholds
- **Precision, Recall, and F1-Score**: Computed separately for safe and unsafe classes to assess class-specific performance

All metrics were computed on the separate test set to provide unbiased estimates of generalization performance.

2.5.5 Visualization

For each embedding model, we created a two-dimensional scatter plot showing:

- All evaluation points colored by ground-truth label (blue for safe, red for unsafe)
- Computed safe and unsafe centroids marked with large X symbols
- Reference points marked with diamond symbols
- Points plotted along the x-axis with y-coordinates indicating perpendicular distance to reference axis

These visualizations provide intuitive insight into how well each model separates the two classes along the reference-defined axis, and whether clear clusters emerge around the centroid positions.

2.5.6 Comparative Analysis

By evaluating three different classifiers on both 2D projections and full embeddings, we can assess (1) whether the 2D projection captures most of the essential information, (2) whether non-linear decision boundaries improve performance (by comparing logistic regression vs. neural networks on full embeddings), and (3) which embedding models provide the most separable representations across different classification paradigms.

This multi-faceted evaluation strategy provides a comprehensive assessment of both the embedding quality and the geometric structure of the ethical distinction in the embedding space.

2.6 Collecting Dataset of Claims in Literature

We procured the academic papers through a web scraping mechanism that filtered through publicly available academic websites such as arXiv and downloaded free PDFs. Our process was as follows:

- Use SemanticScholar and arXiv API to filter for papers contain at least one of fifteen query words related to AI safety (ex: ["AI safety", "AI alignment", "alignment problem", "robustness machine learning", "interpretability AI", "reward hacking", "corrigibility AI", ...])
- Leverage the standard research paper format conventions to filter for conclusion section by setting start index to detect key words such as "Conclusion", "Results", or "Discussion and Conclusion" and ending at key words such as "References" or "Acknowledgements"
- Store conclusions in a jsonl file along with the year of publication for future embedding purposes

While the primary focus of this paper did not use this data, we anticipate using this dataset as one of the primary data sources for our long term project focused on detecting misinformation in real-life scenarios.

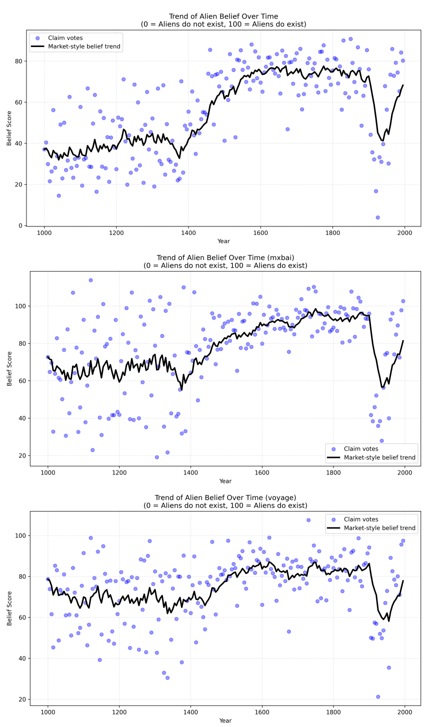
3 Results

3.1 The Projection Method demonstrates Ability to Filter by topic and Order by Semantic Content

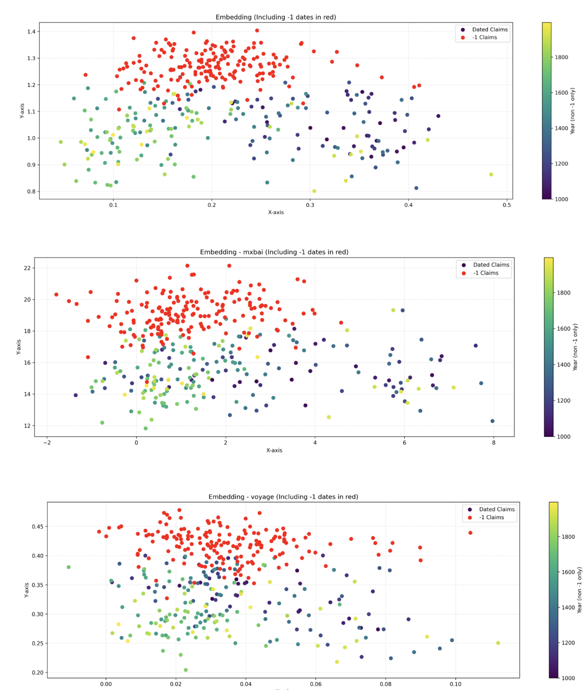
For both the claims on smoking and the claims on the existence of aliens, we observe strong segregation along the perpendicular axis as expected. In particular, for each model, it appears possible to define a threshold wherein a majority of relevant (Smoking or Alien related) claims fall within the threshold, and a majority of irrelevant claims fall outside. For instance, for OpenAI's embedding model, this value falls between 1.1 and 1.2, as can be seen in the top right plot of Figure 1. This value naturally differs between models as a result of different vector embedding magnitudes, and to a lesser extent, the number of dimensions present, which also affects the perpendicular projection distance. Regardless, across all models, we observe strong separation of relevant and irrelevant points.

In addition, we also observe that the opinion of Alien existence across time, as depicted in the first column of figures is correctly represented by the models. In particular, the LLM which generated the claims was instructed to start off with uncertainty, move towards greater confidence in extraterrestrial life, before a sudden drop-off in the 1900s before a sudden recovery. Indeed we find the projection onto the reference axis for all three embedding models is able to track this induced trend.

We find similar results for the experimental design on smoking claims, observing strong separation of smoking-relevant and irrelevant claims, as well as a trend away from the claim that smoking is beneficial for one's health as time progresses, aligning with the truth that is conveyed through the LLM's instructions. Figures for the Smoking experiment can be found in Appendix Figure 2.



(a) Trend over time along reference axis



(b) Plot of candidate claim projections

Figure 1: Projection Methodology applied to claims regarding Aliens. Plots on the left are filtered to only show relevant (alien-related) claims over time; the x-axis represents the LLM assigned year and the y-axis represents the projection onto the reference axis. Plots on the right show all claims plotted with respects to their projection. The x-axis represents the projection along the reference axis and the the y-axis represents the perpendicular projection distance. Each row is a different embedding model, from top to bottom: OpenAI Small, mxbai-embed, and Voyage Lite respectively

3.2 The Projection Method Demonstrates the Ability to Sensitively Classify Highly Semantically Similar Claims Comparable to Traditional Methods

Applying logistic regression to the 2D projection yielded accuracy scores of 60-68% across three embedding models. This is substantially above the 50% random baseline but indicating only modest separability. We find similar performance with the centroid classification scheme, which is computationally simpler than the logistic regression.

We compared our method's performance against two traditional methods: UMAP-based dimensionality reduction and multi-layer perceptron (MLP) classifiers on full embeddings. As we expected, the MLP substantially outperformed our approach (90% vs. 67% accuracy), confirming that our simplistic two-point reference system is unable to fully capture the full nature of ethical claims. However, our method provides interpretability and computational efficiency advantages, as no training phase is required, predictions are O(n) complexity, and results can be directly visualized.

In particular, our reference-based projection (67% accuracy) substantially outperformed UMAP 2D projection with logistic regression (49% accuracy) across all models and metrics. This suggests that a carefully chosen reference axis aligned with the classification objective captures more task-relevant information than general-purpose dimensionality reduction techniques like UMAP, which prioritize preserving variance in the data rather than specific discriminative features.

Table 1: Comprehensive classification performance across three embedding models and four classification approaches. Results show accuracy and AUC-ROC on held-out test set (25% of data, n=65). All models trained with random seed 42 for reproducibility.

	LR (2D Proj.)		**LR (UMAP 2D)**		**MLP (Full Emb.)**		**Centroid (2D)**	
Model	**Acc**	**AUC**	**Acc**	**AUC**	**Acc**	**AUC**	**Acc**	**AUC**
OpenAI Small	0.662	0.720	0.554	0.527	**0.908**	**0.968**	0.631	—
mxbai-embed	**0.738**	**0.778**	0.477	0.481	**0.908**	0.954	0.662	—
Voyage Lite	0.600	0.713	0.446	0.459	0.892	0.925	0.692	—
Mean	*0.667*	*0.737*	*0.492*	*0.489*	*0.903*	*0.949*	*0.662*	—

Table 2: Class-specific performance metrics across all models and classifiers. Bold values indicate best performance within each metric-classifier combination.

		OpenAI Small			**mxbai-embed**			**Voyage Lite**		
Method	**Class**	**P**	**R**	**F1**	**P**	**R**	**F1**	**P**	**R**	**F1**
LR (2D)	Unsafe	0.66	0.70	0.68	**0.74**	**0.76**	**0.75**	0.68	0.39	0.50
	Safe	0.67	0.62	0.65	**0.74**	0.72	**0.73**	0.57	**0.81**	**0.67**
LR (UMAP)	Unsafe	**0.58**	0.42	**0.49**	0.48	**0.45**	0.47	0.46	**0.48**	0.47
	Safe	**0.54**	**0.69**	**0.60**	0.47	0.50	0.48	0.43	0.41	0.42
MLP (Full)	Unsafe	**0.94**	**0.88**	**0.91**	**0.94**	**0.88**	**0.91**	0.91	**0.88**	0.89
	Safe	**0.88**	**0.94**	**0.91**	**0.88**	**0.94**	**0.91**	**0.88**	0.91	**0.89**
Centroid	Unsafe	0.62	**0.70**	0.66	0.64	**0.76**	**0.69**	**0.68**	**0.76**	**0.71**
	Safe	**0.64**	0.56	0.60	**0.69**	0.56	0.62	**0.71**	0.62	**0.67**

Table 1 presents the overall classification performance across all models and methods. Unsurprisingly, the MLP classifier using full embeddings substantially outperforms dimensionality-reduced approaches, achieving over 90% accuracy across all three models. However, the accuracy and AUC-ROC of the 2D projection method we used outperforms the UMAP projection in all cases. We can observe this visually by comparing Supplemental Figure 3 and the results of Supplemental Figure 4; Supplemental Figure 3 portrays a perceivable distinction between centroids of safe and unsafe clusters, while Supplemental Figure 4 indicates significant overlap between safe and unsafe embeddings. The poor performance of traditional techniques for clustering when labeled training data is not present can be explained by the fact that the difference in topic across the data is the primary source of variance, not the difference in safe or unsafe wording. For one to be able to extract such

Table 3: Summary of key findings from comprehensive classifier evaluation. Values represent test set performance (n=65 for supervised methods, n=258 for centroid classifier).

Finding	Evidence
1. Full embeddings critical	MLP using full embeddings achieves 90%+ accuracy (mean 0.903), vastly outperforming dimensionality reduction approaches (2D: 0.667, UMAP 2D: 0.492). This indicates ethical distinctions require access to high-dimensional embedding structure.
2. UMAP loses information	UMAP 2D projections perform worse than 2D reference-based projections (0.492 vs 0.667 accuracy), suggesting the reference-based axis better captures safety-relevant information than general-purpose dimensionality reduction.
3. Model consistency	All three models achieve similar MLP performance (OpenAI & mxbai: 0.908, Voyage: 0.892), with AUC scores 0.925–0.968, indicating comparable quality in the full embedding space despite different architectures.
4. 2D projection viable	Simple 2D projection with logistic regression achieves 60–74% accuracy, comparable to or exceeding centroid baseline (63–69%), demonstrating that a single reference-based axis captures substantial discriminative information.
5. Non-linear boundaries matter	The gap between logistic regression (66.7%) and MLP (90.3%) on the same data indicates that safety distinctions require complex, non-linear decision boundaries in embedding space.

insights without labeled datasets, methods that can extrapolate a small amount of human input into a concrete reference point, such as the proposed projection methodology are crucial.

Our ranking analysis (Table 4) shows that model performance varies by classifier type, with mxbai-embed performing best on 2D projections (0.677) while OpenAI and mxbai tie on the full embedding MLP approach (0.908). Table 3 summarizes our five major findings from this comprehensive evaluation, highlighting the critical importance of preserving full embedding dimensionality for safety classification tasks.

4 Discussion and Conclusion

4.0.1 Discussion

In this paper, we developed and evaluated a simple, interpretable projection technique that uses orthogonal projection to translate high-dimensional vector embeddings onto a 2D graph defined by two user-provided reference points. This approach enables classification of embeddings relative to user-defined semantic dimensions (e.g. safe vs. unsafe ethical content) while providing interpretable visualizations of the embedding space geometry.

We began with proof-of-concept experiments on two datasets (health effects of smoking and existence of aliens) to demonstrate that the projection is able to maintain semantic differences across different topics, and hence, filter claims by topic. The 2D visualizations revealed that claims unrelated to the reference points appear farther from the reference axis (higher y-coordinate), confirming that the orthogonal projection distance correctly captures semantic distance from the reference dimension.

Subsequently, we evaluated this technique's efficacy on a more challenging task: distinguishing between semantically similar claim pairs that share similar words, sentence structure, and topic but differ substantially in ethical implications. We deliberately avoided fine-tuning to assess the capabilities of off-the-shelf, publicly accessible models that could be subject to misuse. We find that the technique greatly outperforms traditional data-unaware techniques such as UMAP, but nonetheless fall short of a full neural-network model that is trained on labeled data and has full access to the entire vector embedding. This is of course unsurprising, as the orthogonal projection necessarily sacrifices some information in return for a significant reduction in computational complexity.

4.0.2 Limitations and Future Work

Due to the limited time capacity of this project, the results exhibited above only evaluate a few aspects of this proposed method. The most primary factor that limits our evaluation is a lack of proper examination of reference points. The reference points used in all experiments were arbitrarily determined, representing a first-pass best guess of human intuition. We find that impressively, even with such crude reference points, the projection method outperforms traditional methods that do not involve training on data. With a more systematic method of determining reference points, it may certainly be possible to greatly bolster the efficacy of the projection method, bringing it even closer to neural networks trained on labeled data.

Next, we note that all datasets are small and generated by LLMs. It may be possible the this projection technique is best suited for LLM generated text, as it could be argued as simply re-projecting the text back into the latent space it was generated from. Although we mitigate this possibility by using human-crafted reference points and embedding models across providers, it would nonetheless be beneficial to test on human text, which is already planned for the long-term continuation of this project, with initial data curation steps covered in the methodology. However, we note that even if the projection methodology is only limited to LLM generated text, it can still have profound impacts on filtering and classifying misinformation in LLM outputs, as its computational efficiency is vastly beyond traditional neural network methods, albeit at lower accuracy levels.

In terms of accuracy levels, given the highly efficient nature of the projection methodology, it would be of great interest to determine whether the mixture of various reference points could be of use. Multiple different projections with multiple different (but related) reference points could be used in a vote-based system to potentially improve accuracy. In particular, given the computationally efficient nature of the method, it could even be possible to utilize a small set of training data to identify optimal reference points. It would hence be of great interest to explore use cases where data is not unavailable, but still sufficiently limited that traditional neural-net models would not be able to train and properly generalize on the data.

Finally, one of the most interesting possibilities this projection methodology opens up is scalability to an arbitrary number of reference points. As mentioned in the methodology, orthogonal projections can be applied to any hyperplane, and defining n reference points enable a projection into a $n-1$ dimension hyperplane, alongside a perpendicular measure. It would be of great interest to test the ability of the model to classify beyond simply 2-reference point binary classifications. One could imagine, for instance, news and media classification that would employ the projection method with three references, for Scientific Advancements, Political/Economic policy, and Fashion/Acting/Pop-culture, for instance. So long as the reference points can be clearly defined, the projection method can be applied, and results easily interpreted. Although, of course, with more reference points, the projection plot would naturally no longer be 2-dimensional.

4.0.3 Conclusion

Although time pressures have created numerous limitations which have inhibited our ability to fully evaluate the potential of the projection method, we nonetheless draw attention to a computationally simple method that does not require labeled training data, but is nonetheless able to cluster and categorize semantic information via vector embeddings. While a majority of embedding interpretation techniques aim to leverage polynomial and exponential-time algorithms on large swaths of labeled training data in hopes for accurate results, we propose a linear-time methodology perfectly suited for when labeled training data and computation are severely limited.

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Appendix and Supplemental Figures

4.1 Mathematical Details of the Projection Method

For two reference vectors $\mathbf{r}_\mathrm{A}$ and $\mathbf{r}_\mathrm{B}$, any candidate embedding $\mathbf{e}_i$ in the dataset, we computed its projection onto the vector formed by the endpoints of $\mathbf{r}_\mathrm{A}$ and $\mathbf{r}_\mathrm{B}$ (mathematically a subtraction of $\mathbf{r}_\mathrm{A}$ and $\mathbf{r}_\mathrm{B}$) using the following geometric construction. Let $d_{12} = \|\mathbf{r}_\mathrm{A} - \mathbf{r}_\mathrm{B}\|_2$ represent the distance between reference points. For each embedding $\mathbf{e}_i$, we computed:

$$d_1 = \|\mathbf{e}_i - \mathbf{r}_\mathrm{A}\|_2, \quad d_2 = \|\mathbf{e}_i - \mathbf{r}_\mathrm{B}\|_2 \tag{3}$$

The one-dimensional coordinate x_i along the reference axis was then calculated using the law of cosines projection:

$$x_i = \frac{d_1^2 - d_2^2 + d_{12}^2}{2 \cdot d_{12}} \tag{4}$$

This formula projects each point onto the line segment between the two reference points, where $x = 0$ corresponds to the position of $\mathbf{r}_\mathrm{A}$ and $x = d_{12}$ corresponds to $\mathbf{r}_\mathrm{B}$. With respects to categorization points with $x_i < d_{12}/2$ are considered to represent $\mathbf{r}_\mathrm{A}$, while those with $x_i > d_{12}/2$ are considered to represent $\mathbf{r}_\mathrm{B}$. For instance, take $\mathbf{r}_\mathrm{A}$ as a statement representing ethical and true facts and $\mathbf{r}_\mathrm{B}$ as a statement representing unethical, dangerous, or misleading facts.

4.2 Details on experimental setup

All experiments were conducted using Python 3.12 with the following key libraries: scikit-learn 1.7.2 for classification algorithms, numpy 1.26.4 for numerical operations, and matplotlib 3.9.2 for visualization. For each embedding model, we used the official API or implementation provided by the model creators to ensure consistency and reproducibility.

4.3 Prompts and reference embeddings for Topic Filtering and Semantic Ordering Experiments

4.3.1 Two reference points embedded to reference vectors to execute the projection methodology on Smoking candidate claims

Smoking is detrimental to human health.

Smoking is beneficial to human health.

4.3.2 Prompt for generating Smoking claims

Return a jsonl object where each entry is of the form "Date": some date, "Claim": some claim, which corresponds with how society's understanding of smoking changed over time. Start with old time claims that smoking is good for relaxing and one's health before the claims shift to the actual truth of smoking causing cancer. You should generate 50 such entries spanning across the ages that reflect the change in belief about smoking. You should make the change slow in some eras and fast in others (not a linear progression), reflecting what actually happened in history

4.3.3 Prompt for generating claims irrelevant to Smoking

Generate a jsonl list of 50 claims of the following form: "Date": "-1", "Claim": "" The date should always be -1, and the claim should be on a variety of topics (politics, truths about the physical world, opinions, etc.) However, the claims should not be about smoking at all.

4.3.4 Two reference points embedded to reference vectors to execute the projection methodology on Alien candidate claims

Extraterrestrial life absolutely exists

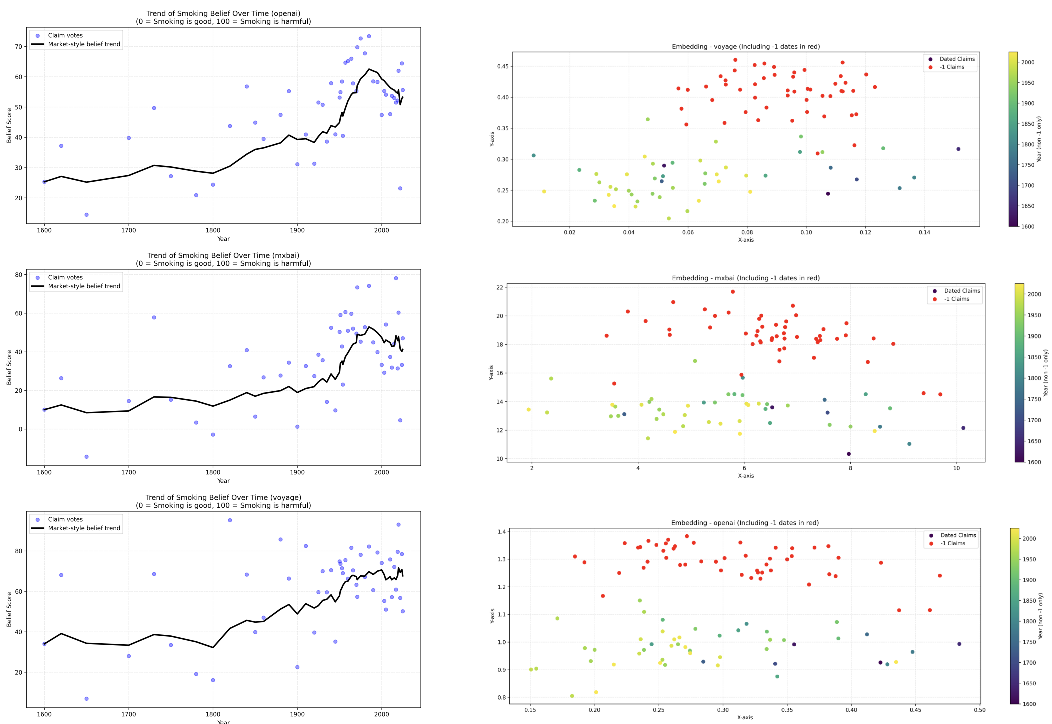
Extraterrestrial life absolutely does not exist

4.3.5 Prompt for generating Alien claims

Return a jsonl object where each entry is of the form "Date": some date, "Claim": some claim, which corresponds with how society's understanding of the existence of aliens has changed over time. Start with old time claims that humans are the center of the universe and that all life exists on earth. You should generate 200 such entries spanning across the ages that reflect the change in belief about aliens. You should make the change slow in some eras and fast in others (not a linear progression), reflecting what actually happend in history. In particular, there should be an era in the 1900s where it is generally concluded that aliens do not exist, before a combination of scientific and conspiracy theory claims suggest life in the universe (ie: science can focus on the possibility of life, and conspiracy can focus on aliens running the government or something)

4.3.6 Prompt for generating claims irrelevant to Aliens

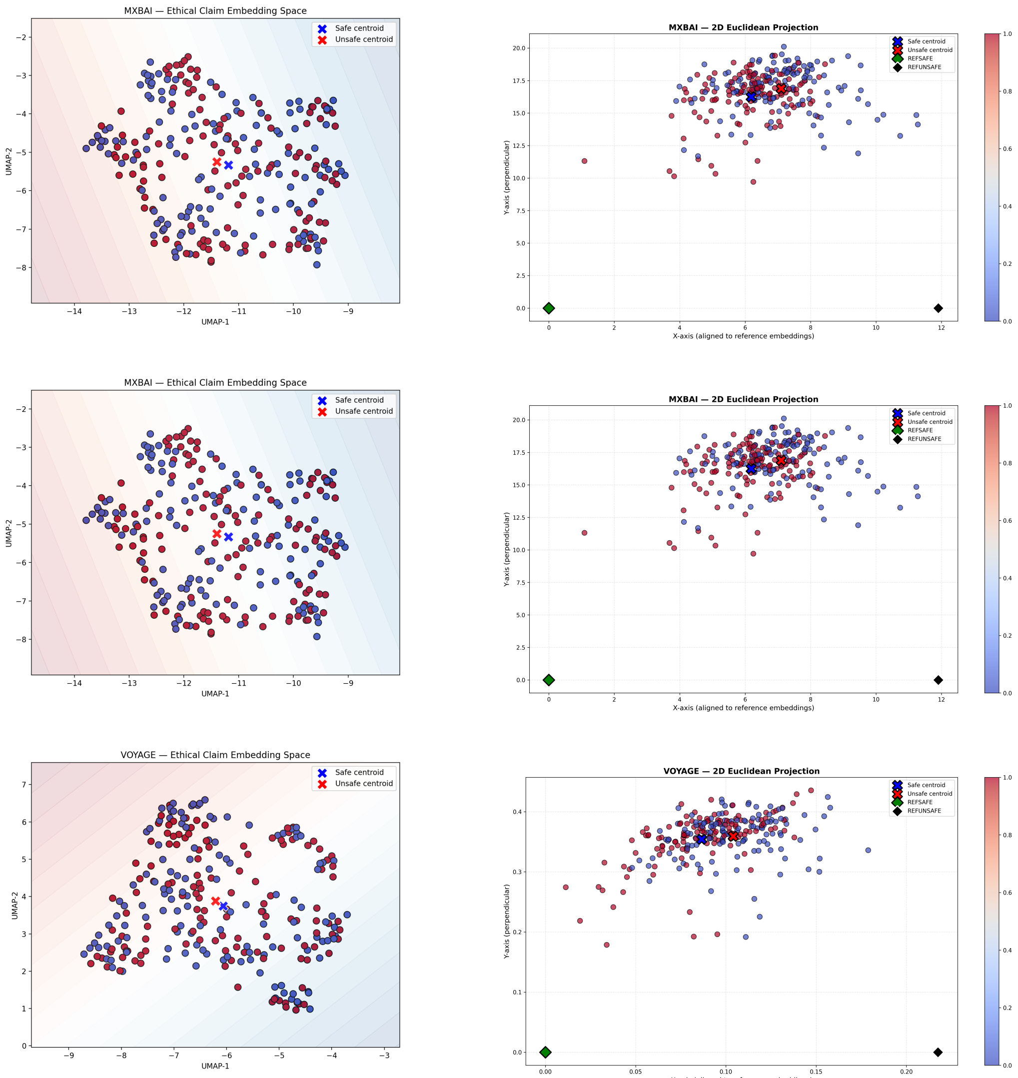
Generate a jsonl list of 50 claims of the following form: "Date": "-1", "Claim": "" The date should always be -1, and the claim should be on a variety of topics (politics, truths about the physical world, opinions, etc.) However, the claims should not be about aliens, life on other planets or alien related conspiracies at all. Feel free to include other conspiracy claims though (ie moon landing, flat earth, anti-vaxx, etc.)



(a) Trend over time along reference axis

(b) Plot of candidate claim projections

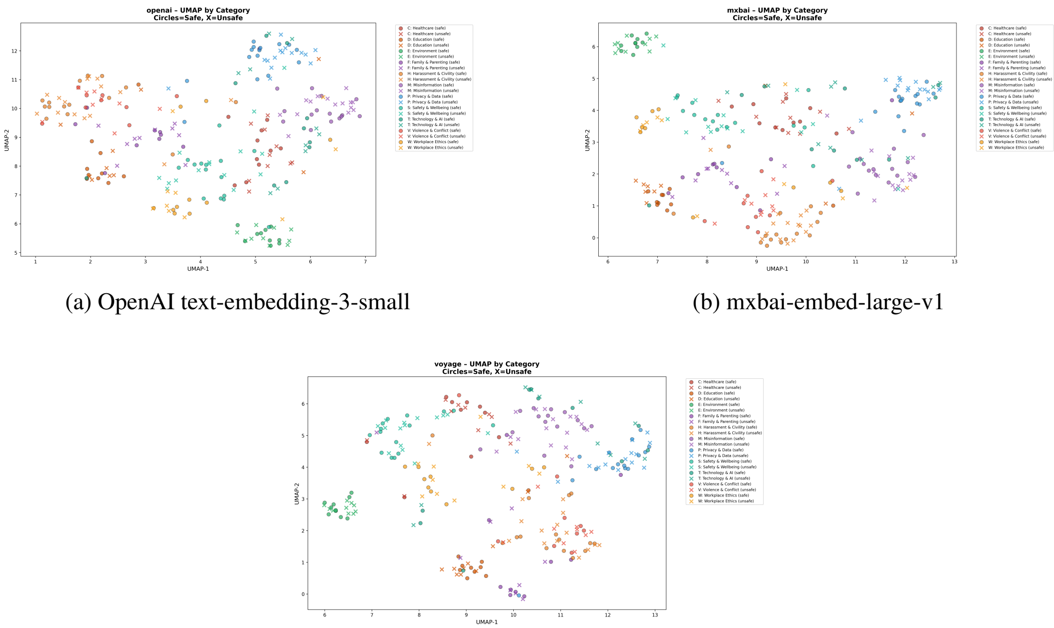
Figure 2: Projection Methodology applied to claims regarding Smoking. Plots on the left are filtered to only show relevant (smoking-related) claims over time; the x-axis represents the LLM assigned year and the y-axis represents the projection onto the reference axis. Plots on the right show all claims plotted with respects to their projection. The x-axis represents the projection along the reference axis and the the y-axis represents the perpendicular projection distance. Each row is a different embedding model, from top to bottom: OpenAI Small, mxbai-embed, and Voyage Lite respectively



(a) Claims clustered by UMAP

(b) Claims clustered by Projection Method

Figure 3: Embedding representations for the UMAP and Projection method, on the left and right respectively. (Top) OpenAI text-embedding-3-small, (Middle) mxbai-embed-large-v1, (Bottom) Voyage AI voyage-lite-02-instruct. Blue circles represent safe claims, red circles represent unsafe claims. The X markers indicate class centroids, while diamond markers show reference points. Note: due to code versioning differences the vector embeddings used in the projection and UMAP graphs are different, although the underlying text prompt remains the same.



(a) OpenAI text-embedding-3-small

(b) mxbai-embed-large-v1

(c) Voyage AI voyage-lite-02-instruct

Figure 4: UMAP projection visualizations comparing embedding space organization across three models, color-coded by topic. Each plot displays safe (circles) and unsafe (X symbols) claims projected onto a 2D space. Embedding space can be visibly grouped by topic similarity (colors) but significantly overlaps between safe/unsafe embeddings (circles and X symbols).

Table 4: Ranking of embedding models by accuracy within each classifier type. Rankings reveal model strengths vary by classification approach.

Classifier	Rank	Model	Accuracy
LR (2D)	1st	mxbai-embed	0.738
	2nd	OpenAI Small	0.662
	3rd	Voyage Lite	0.600
LR (UMAP)	1st	OpenAI Small	0.554
	2nd	mxbai-embed	0.477
	3rd	Voyage Lite	0.446
MLP (Full)	1st	**OpenAI & mxbai**	**0.908**
	2nd	Voyage Lite	0.892
	—	—	—
Centroid	1st	Voyage Lite	0.692
	2nd	mxbai-embed	0.662
	3rd	OpenAI Small	0.631